

4	(a)		<u>work done per unit positive charge</u> in bringing a small <u>test charge</u> from <u>infinity to that point</u>	B1
	(b)	(i)	p.d. = 15 V	A1
		(ii)	change in potential $\Delta V = 25 - 5 = 20 \text{ V}$ work done = $q\Delta V = 5 \times 10^{-6} \times 20$ $= 1.0 \times 10^{-4} \text{ J}$	M1 A1
		(iii)	The potential increase positively as the charge move from B to D. Since it is a positive charge, work has to be done by external agent to move it to a more positive potential line.	B1 B1
	(c)	(i)	electric field strength at a point is numerically equal to the potential gradient at that point	B1
		(ii)	towards the +20 V equipotential line and approximately perpendicular at the point.	B1
		(iii)	move away to the right / towards lower potential / away from higher potential initial acceleration is small, acceleration is (much) larger beyond the +20 V line	B1 B1
				[Total: 10]